ASSIGNMENT 1 - LINUX BASICS

1- Introduction to linux

Linux is an **open-source operating system** (OS) that helps your computer run and manage hardware and software—just like Windows or macOS.

- Linux is a **free and open-source OS**
- It was created by **Linus Torvalds** in 1991
- Anyone can **use, modify, and share** it

🔹 Key Features of Linux

- **Free to use** 💰
- **Secure** 🔒 (less virus compared to Windows)
- **Multi-user** 👥 (many users can use at same time)
- **Multitasking** ⚙️ (run many programs together)
- **Highly customizable** 🎨

🔹💻 Where is Linux Used?

- Servers (Google, Facebook use Linux)
- Mobile phones (Android is based on Linux)
- Supercomputers
- Personal computers

⚙️ Basic Components of Linux

- **Kernel** 🔹
  - Core part of Linux
  - Controls hardware (CPU, memory)
- **Shell** 💬
  - Interface where you type commands
  - Example: Terminal
- **File System** 📂
  - Organizes files and folders
- **Applications** 📱
  - Software like browser, text editor

2-Linux History and Philosophy

 1960s – UNIX started

- An operating system called **UNIX** was created in research labs
- It became the base idea for Linux

 1980s – GNU Project began

- **Richard Stallman** started the GNU Project
- Goal: Create a **free operating system**
- Many tools were created, but **kernel was missing**

 1991 – Linux created

- **Linus Torvalds** (a student from Finland) created Linux kernel
- He shared it freely on the internet

 1992 – Linux became open-source

- Linux was released under **GPL license**
- Anyone could use, modify, and share it

 1990s – Growth of Linux

- Developers around the world improved Linux
- Different versions (distributions) like Ubuntu, Fedora started coming

 2000s – Popularity increased

- Linux started being used in:
  - Servers
  - Companies
  - Supercomputers

 Today – Everywhere

- Linux is used in:
  - Android phones
  - Cloud computing
  - Big companies like Google

3- Popular Linux Distributions

👉 **Linux distributions (distros)** are different versions of Linux.
All use the same Linux core (kernel), but they differ in **look, features, and usage**.

1. Ubuntu

- Most popular and **beginner-friendly**
- Easy to install and use
- Has a **simple graphical interface (GUI)**
- Strong community support

Used for: Students, beginners, daily use

2. Fedora

- Developed with support from **Red Hat**
- Comes with **latest features and updates**
- Slightly advanced compared to Ubuntu

Used for: Developers, learning new technologies

3. CentOS

- Stable and secure
- Based on Red Hat Enterprise Linux (RHEL)
- Not updated frequently (focus on stability)

Used for: Servers, companies

4. Red Hat Enterprise Linux (RHEL)

- Paid version of Linux
- Very secure and stable
- Provides **official support**

Used for: Big companies, enterprise systems

5. Debian

- One of the **oldest and most stable** distros
- Not very flashy but very reliable
- Base for many other distros (like Ubuntu)

4-Understanding Desktop Environment (GNOME, KDE, XFCE)

What is a Desktop Environment?

- A **Desktop Environment (DE)** is what you **see and use on screen** in Linux

It includes:

- Windows ▯
- Icons 🗀
- Menus 📋
- Taskbar

🌟 1. GNOME

◈ What is GNOME?(GNU Network Object Model Environment)

- A **simple and modern desktop environment**
- Default in many Linux distros like Ubuntu

◈ Features

- Clean and minimal design ✨
- Easy to use (good for beginners)
- Focus on simplicity (less clutter)

◈ Example

- Looks like a **mobile-style interface** with apps and search

👉 **Best for:** Beginners and users who like simple UI

🎨 2. KDE

◈ What is KDE?(K Desktop Environment)

A **powerful and highly customizable desktop environment**

◈ Features

- Very colorful and attractive 🌈
- You can change almost everything (themes, icons, layout)
- More features than GNOME

◈ Example

- Similar to Windows (taskbar, start menu)

⚡ 3. Xfce

◈ What is XFCE?(XForms Common Environment)

- A **lightweight and fast desktop environment**

◈ Features

- Uses very less RAM 💨
- Simple and basic interface
- Works well on old/low-performance computers

◈ Example

- Looks simple like older Windows versions

👉 **Best for:** Old PCs, fast performance

5- Command Line Basics in Linux

💡 What is Command Line?

- The **command line** is a place where you **type commands to control the computer**
- In Linux, it is also called **Terminal**

👉 Simple:
Instead of clicking with mouse, you type commands to do work.

📟 What is Terminal?

- Terminal is an application where you type commands
- It uses a program called **Bash** (a shell)

▯ What is Shell?

- A **shell** is a program that:
  - Takes your command
  - Sends it to the system (kernel)
  - Shows output

👉 Example shells: Bash, Zsh

🗁 Basic Concepts

1. Directory (Folder)

- A place to store files
- Example: Documents, Downloads

2. Path

- The **location of a file or folder**
- Example: /home/user/Documents

3. Root Directory

- Top level directory in Linux
- Represented by /

🔑 Important Basic Commands

🗁 1. ls

- Shows list of files/folders
  ☞ Example: ls

🗁 2. cd

- Change directory (move between folders)
  ☞ Example: cd Documents

🔑 3. pwd

- Shows current location
  ☞ Example: pwd

🗎 4. touch

- Creates a new file
  ☞ Example: touch file.txt

🗏 5. mkdir

- Creates a new folder
  ☞ Example: mkdir myfolder

✕ 6. rm

- Deletes files/folders
  ☞ Example: rm file.txt

🗏 7. cp

- Copies files
  ☞ Example: cp file.txt backup.txt

🔁 8. mv

- Moves or renames files
  👉 Example: mv file.txt newname.txt

🧹 9. clear

- Clears terminal screen

⚙️ Command Structure

- 👉 Most Linux commands follow this format:
  `command [options] [arguments]`
- Example:
  `ls -l /home`

6- Terminal Fundamentals in Linux

💡 What is Terminal?

- The **Terminal** is a program where you **type commands to control Linux**
- It is also called **Command Line Interface (CLI)**

  👉 Simple:
  Terminal = place to type commands instead of using mouse

🔄 How Terminal Works

1. You type a command ⌨️
2. The **Bash** (shell) reads it
3. It sends it to the Linux system (kernel)
4. Output is shown on screen

💻 Accessing the Terminal in Linux

💡 What does "Accessing the Terminal" mean?

- It means **opening the Terminal window** so you can type commands and control Linux

Ways to Open Terminal

1. **Using Keyboard Shortcut (Fastest)**
   - Press: **Ctrl + Alt + T**

👉 This opens the terminal instantly in most Linux systems

2. **From Application Menu**
   - Click on **Menu / Applications**
   - Search for **"Terminal"**
   - Click on it to open
3. **Right Click Method**
   - Right-click on desktop or inside a folder
   - Click **"Open Terminal"** or **"Open in Terminal"**

4. **Using Run Command**
   - Press: **Alt + F2**
   - Type:
     gnome-terminal
   - Press Enter

👉 Works in systems using **GNOME**

🗁 7. Directory Structure in Linux

👉 Linux uses a **tree-like structure** (like folders inside folders)

- The top-most directory is called **Root (/)**
- All files and folders start from here

Important Directories

🔹 Root Directory (/)

- Top level of Linux
- Everything starts from here

👉 Simple: **Main folder of system**

🔹 Home Directory (/home)

- Stores **user files (documents, photos, etc.)**
- 👉 Example:- /home/aastha(For Personal folder)

🔹 Bin Directory (/bin)

- Contains **basic system commands** like ls, cp, mv

👉 Uses: **Commands storage folder**

🔹 Other Important Directories

- /etc → system settings ⚙

- /usr → software and programs 📦
- /var → logs and variable data 📄
- /tmp → temporary files 🗑️

Absolute vs Relative Paths

🔷 Absolute Path

Full path from root /

👉 Example: /home/aastha/Documents

👉 Uses: **Complete address**

🔷 Relative Path

- Path based on current location

👉 Example: cd Documents

👉 Simple: **Short/local address**

🗂️ File Management Commands

👉 These commands help you **create, view, and manage files/folders**

✏️ Creating Files & Directories

📄 Create File

touch file.txt

🗂️ Create Folder

mkdir myfolder

📋 Listing Files

ls

Options:

- ls -l → detailed view
- ls -a → hidden files

❌ Deleting Files & Directories

🗑️ Delete File

```
rm file.txt
```

🗁 **Delete Folder**

```
rm -r myfolder
```

🗎 **Copy Files**

```
cp file.txt backup.txt
```

🔄 **Move / Rename Files**

```
mv file.txt newname.txt
```

📝 One Line Summary

👉 **Linux directory structure organizes files, paths help locate them, and commands are used to manage files and folders.**